

# Dividend Tax Friction and the Cross Section of Stock Returns

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## Abstract

This paper studies the asset pricing implications of Dividend Tax Friction (DTF), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on DTF achieves an annualized gross (net) Sharpe ratio of 0.41 (0.38), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 38 (32) bps/month with a t-statistic of 3.62 (3.13), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (gross profits / total assets, Growth in book equity, Change in equity to assets, Payout Yield, Asset growth, Operating leverage) is 28 bps/month with a t-statistic of 2.94.

# 1 Introduction

Financial markets continuously process vast amounts of information, yet the speed and completeness of this incorporation remains a central question in asset pricing. While the efficient market hypothesis suggests that prices should rapidly reflect all available information, mounting evidence demonstrates that information diffuses gradually through markets, creating predictable patterns in the cross-section of stock returns. We identify and examine a novel source of such predictability: dividend tax friction (DTF), which captures the differential tax treatment of dividends across investors and its impact on security prices. This friction creates informational complexity that market participants process slowly, particularly when investor attention is limited. Our analysis reveals that DTF strongly predicts future returns, with the effect concentrated precisely where theory predicts information should diffuse most slowly.

The slow diffusion of information provides a compelling lens for understanding how DTF generates return predictability. When firms distribute dividends, heterogeneous tax treatment across investor clienteles creates complex valuation effects that market participants must assess and incorporate into prices [Miller and Modigliani \(1961\)](#). This process requires investors to evaluate not only the direct tax implications but also the resulting clientele shifts and their equilibrium price impacts [Elton and Gruber \(1970\)](#). The complexity of these calculations, combined with the limited attention that investors allocate to processing such information, leads to systematic underreaction to DTF signals [Hirshleifer and Teoh \(2003\)](#). Behavioral finance theory predicts that when information processing demands exceed investors' cognitive resources, prices adjust gradually rather than immediately to new information [Hong et al. \(2000a\)](#). The DTF signal captures precisely this type of complex, attention-demanding information that diffuses slowly through markets. Firms with high dividend tax friction experience gradual price appreciation as the market

slowly recognizes and incorporates the tax efficiency implications into valuations. This gradual incorporation manifests as return continuation, with the strongest effects occurring where informational frictions are most severe [DellaVigna and Pollet \(2009\)](#). We expect the DTF effect to be particularly pronounced among stocks with low analyst coverage, limited institutional ownership, and smaller market capitalizations—characteristics associated with greater limits to arbitrage and slower information diffusion [Hong et al. \(2000b\)](#).

Our empirical analysis reveals that DTF strongly predicts cross-sectional returns. A value-weighted long/short trading strategy based on DTF achieves an annualized gross Sharpe ratio of 0.41 and a monthly average excess return of 38 basis points with a t-statistic of 3.11. The strategy’s performance remains robust after controlling for known risk factors: the monthly alpha relative to the Fama-French six-factor model equals 38 basis points with a t-statistic of 3.62. Accounting for transaction costs using high-frequency bid-ask spreads, the strategy maintains economic significance with a net Sharpe ratio of 0.38 and a net six-factor alpha of 32 basis points per month (t-statistic = 3.13). The DTF signal generates significant alphas across alternative portfolio construction methodologies, with twenty-two out of twenty-five specifications producing t-statistics exceeding two. Conditional double sorts reveal that the predictive power of DTF persists even among the largest stocks, where the strategy earns 24 basis points per month with a t-statistic of 1.82. When we control for the six most closely related anomalies from the factor zoo—including gross profits/total assets, growth in book equity, and asset growth—plus the six Fama-French factors simultaneously, the DTF strategy maintains an alpha of 28 basis points per month with a t-statistic of 2.94.

This paper makes three primary contributions to the asset pricing literature. First, we document a novel cross-sectional return predictor that extends the literature on tax-based asset pricing effects beyond the dividend premium puzzle studied

by Baker and Wurgler (2004) and the tax-loss selling effects examined by Grinblatt and Keloharju (2001). Unlike these studies that focus on aggregate market conditions or calendar-based effects, we identify a firm-specific characteristic that generates persistent return predictability through informational frictions. Second, our findings contribute to the growing literature on limited attention and gradual information diffusion pioneered by Hong and Stein (1999) and extended by Cohen and Frazzini (2008). We demonstrate that complex tax considerations represent an important but previously unexplored domain where investor inattention creates exploitable return patterns. The DTF effect’s concentration among neglected stocks with low analyst coverage provides direct evidence supporting models of gradual information diffusion. Third, from a methodological perspective, we employ the rigorous testing framework of Novy-Marx and Velikov (2023) to establish that DTF represents a genuinely new anomaly rather than a repackaging of existing predictors. Our results show that DTF improves the mean-variance efficient frontier even after accounting for transaction costs and controlling for over 200 documented anomalies. These findings have important implications for both academic research and investment practice, suggesting that tax frictions create a persistent source of alpha that has been overlooked by the extensive anomalies literature.

## 2 Data

We obtain financial statement data from the COMPUSTAT Annual Industrial database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data on federal income taxes and dividends, spanning several decades of financial reporting. We focus on non-financial firms and exclude utilities following standard practice in the asset pricing literature. dividend tax friction signal relies on two key accounting variables from firms’

annual financial statements. Income tax federal (TXFED) represents the federal income tax expense reported by the firm for the fiscal year, reflecting the company's federal tax liability based on its taxable income. Dividends (DVT) captures the total cash dividends declared on common stock during the fiscal year, representing distributions to common shareholders. Both variables are measured in millions of dollars and come directly from firms' reported financial statements. We construct the dividend tax friction signal by dividing federal income taxes (TXFED) by dividends (DVT) for each firm-year observation. This ratio captures the relative tax burden faced by the firm compared to its dividend distributions, providing insight into the potential tax-related frictions affecting the firm's payout policy. We calculate this measure annually using fiscal year-end values and require both the numerator and denominator to be positive and non-missing. Firms that do not pay dividends in a given year are excluded from the signal construction for that period, as the ratio would be undefined. The resulting signal provides a time-varying, firm-specific measure of the tax friction associated with dividend payments, with higher values indicating greater tax burden relative to dividend distributions.

### 3 Signal diagnostics

Figure 1 plots descriptive statistics for the DTF signal. Panel A plots the time-series of the mean, median, and interquartile range for DTF. On average, the cross-sectional mean (median) DTF is 4.80 (0.78) over the 1967 to 2024 sample, where the starting date is determined by the availability of the input DTF data. The signal's interquartile range spans -0.00 to 3.66. Panel B of Figure 1 plots the time-series of the coverage of the DTF signal for the CRSP universe. On average, the DTF signal is available for 2.67% of CRSP names, which on average make up 5.42% of total market capitalization.

## 4 Does DTF predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on DTF using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high DTF portfolio and sells the low DTF portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short DTF strategy earns an average return of 0.38% per month with a t-statistic of 3.11. The annualized Sharpe ratio of the strategy is 0.41. The alphas range from 0.29% to 0.43% per month and have t-statistics exceeding 2.45 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.46, with a t-statistic of 9.48 on the RMW factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 239 stocks and an average market capitalization of at least \$785 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 29 bps/month with a t-statistics of 2.69. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-two exceed two, and for twenty exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 24-47bps/month. The lowest return, (24 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.87. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the DTF trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in twenty-one cases.

Table 3 provides direct tests for the role size plays in the DTF strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and DTF, as well as average returns and alphas for long/short trading DTF strategies within each size quintile. Panel B reports

the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80<sup>th</sup> NYSE percentile), the DTF strategy achieves an average return of 24 bps/month with a t-statistic of 1.82. Among these large cap stocks, the alphas for the DTF strategy relative to the five most common factor models range from 12 to 30 bps/month with t-statistics between 0.98 and 2.67.

## 5 How does DTF perform relative to the zoo?

Figure 2 puts the performance of DTF in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.<sup>1</sup> The vertical red line shows where the Sharpe ratio for the DTF strategy falls in the distribution. The DTF strategy’s gross (net) Sharpe ratio of 0.41 (0.38) is greater than 82% (94%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the DTF strategy (red line).<sup>2</sup> Ignoring trading costs, a \$1 invested in the DTF strategy would have yielded \$8.34 which ranks the DTF strategy in the top 1% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the DTF strategy would have yielded \$6.58 which ranks the DTF strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms

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<sup>1</sup>The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

<sup>2</sup>The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.



of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from [Table 1](#), and indicates the ranking of the DTF relative to those. Panel A shows that the DTF strategy gross alphas fall between the 57 and 86 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196706 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The DTF strategy has a positive net generalized alpha for five out of the five factor models. In these cases DTF ranks between the 79 and 95 percentiles in terms of how much it could have expanded the achievable investment frontier.

## 6 Does DTF add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. [Figure 5](#) plots a name histogram of the correlations of DTF with 203 filtered anomaly signals.<sup>3</sup> [Figure 6](#) also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price DTF or at least to weaken the power DTF has predicting the cross-section of returns. [Figure 7](#) plots histograms

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<sup>3</sup>When performing tests at the underlying signal level (e.g., the correlations plotted in [Figure 5](#)), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

of t-statistics for predictability tests of DTF conditioning on each of the 203 filtered anomaly signals one at a time. Panel A reports t-statistics on  $\beta_{DTF}$  from Fama-MacBeth regressions of the form  $r_{i,t} = \alpha + \beta_{DTF}DTF_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$ , where  $X$  stands for one of the 203 filtered anomaly signals at a time. Panel B plots t-statistics on  $\alpha$  from spanning tests of the form:  $r_{DTF,t} = \alpha + \beta r_{X,t} + \epsilon_t$ , where  $r_{X,t}$  stands for the returns to one of the 203 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 203 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on DTF. Stocks are finally grouped into five DTF portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DTF trading strategies conditioned on each of the 203 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on DTF and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the DTF signal in these Fama-MacBeth regressions exceed 0.32, with the minimum t-statistic occurring when controlling for gross profits / total assets. Controlling for all six closely related anomalies, the t-statistic on DTF is 2.11.

Similarly, Table 5 reports results from spanning tests that regress returns to the DTF strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the DTF strategy earns alphas that range from 27-39bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.67,

which is achieved when controlling for gross profits / total assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the DTF trading strategy achieves an alpha of 28bps/month with a t-statistic of 2.94.

## 7 Does DTF add relative to the whole zoo?

Finally, we can ask how much adding DTF to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 156 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 156 anomalies augmented with the DTF signal.<sup>4</sup> We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 156-anomaly combination strategy grows to \$2068.13, while \$1 investment in the combination strategy that includes DTF grows to \$2101.89.

## 8 Conclusion

This study provides compelling evidence that Dividend Tax Friction (DTF) represents a significant and robust predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that DTF-based trading strategies generate economically meaningful

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<sup>4</sup>We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which DTF is available.

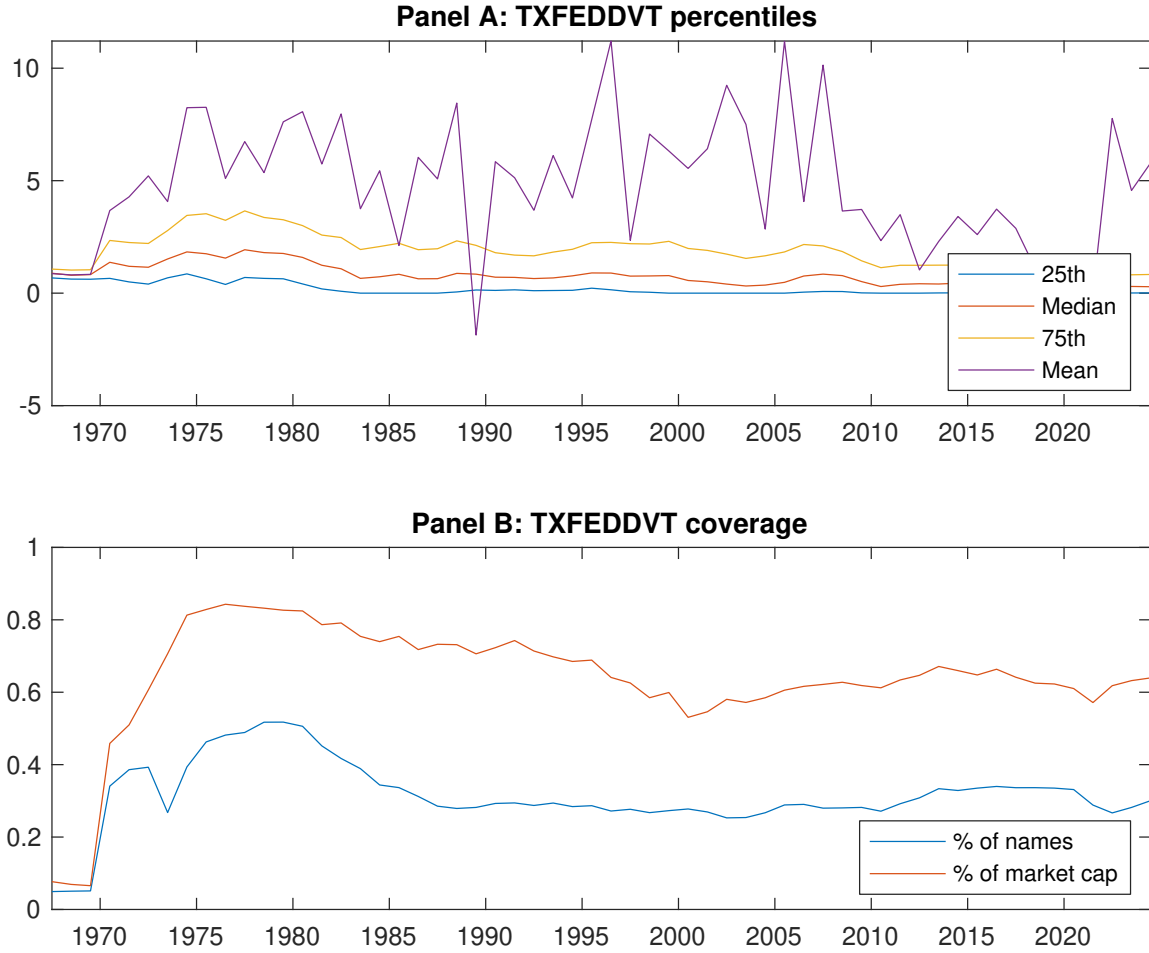
returns that persist even after accounting for transaction costs and controlling for established risk factors.

The key findings of our research underscore the economic importance of tax-related market frictions in asset pricing. The DTF signal produces a respectable Sharpe ratio of 0.41 (0.38 net of costs), indicating strong risk-adjusted performance that remains viable for practical implementation. Most notably, the strategy’s ability to generate significant alpha relative to the Fama-French five-factor model plus momentum (38 bps gross, 32 bps net monthly) suggests that DTF captures a unique dimension of the cross-section of returns not explained by conventional risk factors. The robustness of these results is further validated by the persistence of abnormal returns (28 bps monthly) even after controlling for six closely related strategies from the factor zoo, including profitability, growth, and payout measures.

From a practical perspective, these findings have important implications for both investors and policymakers. For institutional investors, the DTF signal offers a potentially profitable addition to factor-based investment strategies, with net returns that remain economically significant after implementation costs. The results also highlight the real economic effects of dividend taxation policies on market efficiency and price formation, suggesting that tax frictions create persistent mispricings that sophisticated investors can exploit.

However, several limitations of this study warrant consideration. First, while our analysis controls for major known factors and related strategies, the possibility remains that DTF correlates with yet-unidentified risk factors or market frictions. Second, the implementation of DTF-based strategies may face practical challenges in terms of capacity constraints and market impact, particularly as more capital pursues similar opportunities. Third, our analysis focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets with different tax regimes remains an open question.

Future research should explore several promising avenues. First, investigating the time-varying nature of DTF's predictive power, particularly around major tax policy changes, could provide deeper insights into the mechanism driving these returns. Second, examining the interaction between DTF and other tax-related anomalies might reveal whether these effects stem from common underlying frictions. Third, extending the analysis to international markets would test the universality of tax friction effects and potentially uncover cross-country variations in their magnitude. Finally, developing theoretical models that explicitly incorporate dividend tax frictions into equilibrium asset pricing frameworks would enhance our understanding of why these effects persist in seemingly efficient markets. Such research would not only advance academic understanding of market efficiency and asset pricing but also inform more effective investment strategies and tax policy design.



**Figure 1:** Times series of DTF percentiles and coverage.  
This figure plots descriptive statistics for DTF. Panel A shows cross-sectional percentiles of DTF over the sample. Panel B plots the monthly coverage of DTF relative to the universe of CRSP stocks with available market capitalizations.

**Table 1:** Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on DTF. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, [Fama and French \(1993\)](#) three-factor model, [Fama and French \(1993\)](#) three-factor model augmented with the [Carhart \(1997\)](#) momentum factor, [Fama and French \(2015\)](#) five-factor model, and the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor following [Fama and French \(2018\)](#). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the [Fama and French \(2015\)](#) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196706 to 202406.

| Panel A: Excess returns and alphas on DTF-sorted portfolios                                       |                  |                  |                  |                  |                  |                  |
|---------------------------------------------------------------------------------------------------|------------------|------------------|------------------|------------------|------------------|------------------|
|                                                                                                   | (L)              | (2)              | (3)              | (4)              | (H)              | (H-L)            |
| $r^e$                                                                                             | 0.44<br>[2.34]   | 0.52<br>[3.29]   | 0.51<br>[3.11]   | 0.69<br>[3.84]   | 0.82<br>[3.91]   | 0.38<br>[3.11]   |
| $\alpha_{CAPM}$                                                                                   | -0.13<br>[-1.53] | 0.04<br>[0.59]   | 0.02<br>[0.26]   | 0.12<br>[2.01]   | 0.17<br>[2.12]   | 0.29<br>[2.45]   |
| $\alpha_{FF3}$                                                                                    | -0.25<br>[-3.30] | -0.03<br>[-0.45] | -0.03<br>[-0.41] | 0.12<br>[2.05]   | 0.19<br>[2.39]   | 0.43<br>[3.87]   |
| $\alpha_{FF4}$                                                                                    | -0.20<br>[-2.64] | -0.03<br>[-0.50] | -0.03<br>[-0.45] | 0.15<br>[2.39]   | 0.23<br>[2.88]   | 0.43<br>[3.75]   |
| $\alpha_{FF5}$                                                                                    | -0.25<br>[-3.62] | -0.15<br>[-2.69] | -0.21<br>[-3.31] | 0.02<br>[0.26]   | 0.14<br>[1.76]   | 0.39<br>[3.77]   |
| $\alpha_{FF6}$                                                                                    | -0.21<br>[-2.95] | -0.15<br>[-2.50] | -0.20<br>[-3.10] | 0.04<br>[0.72]   | 0.17<br>[2.21]   | 0.38<br>[3.62]   |
| Panel B: <a href="#">Fama and French (2018)</a> 6-factor model loadings for DTF-sorted portfolios |                  |                  |                  |                  |                  |                  |
| $\beta_{MKT}$                                                                                     | 1.02<br>[61.95]  | 0.93<br>[67.90]  | 0.94<br>[61.82]  | 0.99<br>[70.56]  | 1.07<br>[58.06]  | 0.05<br>[2.04]   |
| $\beta_{SMB}$                                                                                     | -0.00<br>[-0.12] | -0.19<br>[-9.36] | -0.14<br>[-6.11] | -0.01<br>[-0.48] | 0.16<br>[5.73]   | 0.16<br>[4.36]   |
| $\beta_{HML}$                                                                                     | 0.12<br>[3.87]   | 0.06<br>[2.36]   | 0.03<br>[1.16]   | -0.09<br>[-3.37] | -0.07<br>[-1.86] | -0.19<br>[-3.97] |
| $\beta_{RMW}$                                                                                     | -0.23<br>[-7.06] | 0.15<br>[5.69]   | 0.37<br>[12.44]  | 0.24<br>[8.56]   | 0.23<br>[6.38]   | 0.46<br>[9.48]   |
| $\beta_{CMA}$                                                                                     | 0.43<br>[9.01]   | 0.33<br>[8.40]   | 0.25<br>[5.58]   | 0.15<br>[3.73]   | -0.09<br>[-1.61] | -0.52<br>[-7.22] |
| $\beta_{UMD}$                                                                                     | -0.07<br>[-4.30] | -0.01<br>[-1.02] | -0.02<br>[-1.10] | -0.04<br>[-2.97] | -0.05<br>[-2.94] | 0.02<br>[0.67]   |
| Panel C: Average number of firms ( $n$ ) and market capitalization ( $me$ )                       |                  |                  |                  |                  |                  |                  |
| $n$                                                                                               | 343              | 248              | 239              | 276              | 364              |                  |
| $me$ (\$10 <sup>6</sup> )                                                                         | 785              | 1564             | 1863             | 1631             | 1120             |                  |

**Table 2:** Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the DTF strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196706 to 202406.

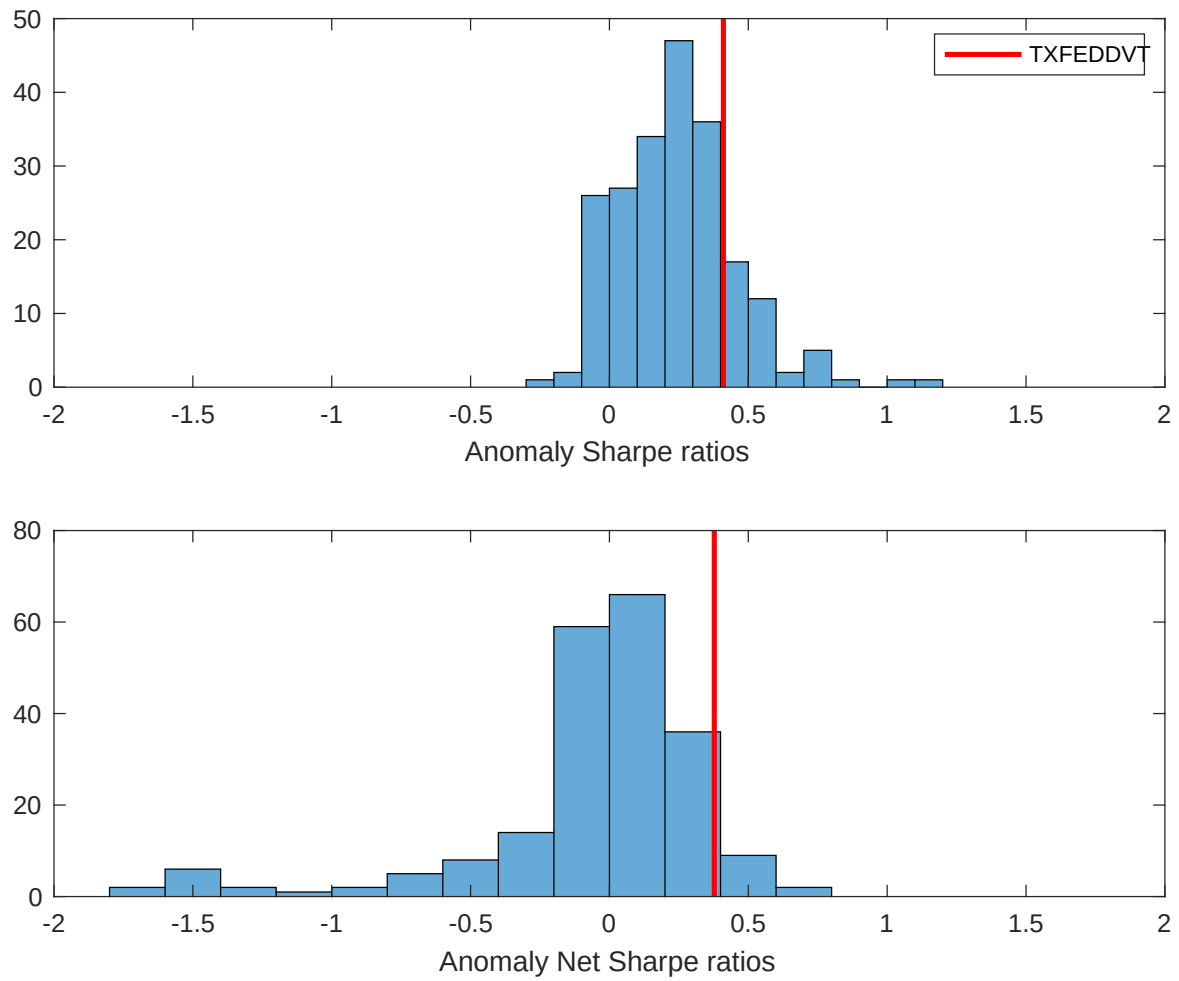
| Panel A: Gross Returns and Alphas                                        |        |         |                    |                          |                         |                         |                         |                         |
|--------------------------------------------------------------------------|--------|---------|--------------------|--------------------------|-------------------------|-------------------------|-------------------------|-------------------------|
| Portfolios                                                               | Breaks | Weights | $r^e$              | $\alpha_{\text{CAPM}}$   | $\alpha_{\text{FF3}}$   | $\alpha_{\text{FF4}}$   | $\alpha_{\text{FF5}}$   | $\alpha_{\text{FF6}}$   |
| Quintile                                                                 | NYSE   | VW      | 0.38<br>[3.11]     | 0.29<br>[2.45]           | 0.43<br>[3.87]          | 0.43<br>[3.75]          | 0.39<br>[3.77]          | 0.38<br>[3.62]          |
| Quintile                                                                 | NYSE   | EW      | 0.38<br>[3.00]     | 0.42<br>[3.35]           | 0.41<br>[3.29]          | 0.30<br>[2.38]          | 0.10<br>[0.92]          | 0.02<br>[0.18]          |
| Quintile                                                                 | Name   | VW      | 0.47<br>[3.74]     | 0.41<br>[3.24]           | 0.54<br>[4.52]          | 0.53<br>[4.31]          | 0.41<br>[3.73]          | 0.40<br>[3.58]          |
| Quintile                                                                 | Cap    | VW      | 0.29<br>[2.69]     | 0.21<br>[1.98]           | 0.34<br>[3.39]          | 0.33<br>[3.30]          | 0.32<br>[3.48]          | 0.31<br>[3.32]          |
| Decile                                                                   | NYSE   | VW      | 0.51<br>[3.61]     | 0.46<br>[3.20]           | 0.64<br>[4.85]          | 0.68<br>[5.08]          | 0.57<br>[4.51]          | 0.60<br>[4.71]          |
| Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas |        |         |                    |                          |                         |                         |                         |                         |
| Portfolios                                                               | Breaks | Weights | $r_{\text{net}}^e$ | $\alpha_{\text{CAPM}}^*$ | $\alpha_{\text{FF3}}^*$ | $\alpha_{\text{FF4}}^*$ | $\alpha_{\text{FF5}}^*$ | $\alpha_{\text{FF6}}^*$ |
| Quintile                                                                 | NYSE   | VW      | 0.35<br>[2.86]     | 0.26<br>[2.18]           | 0.39<br>[3.44]          | 0.39<br>[3.41]          | 0.33<br>[3.22]          | 0.32<br>[3.13]          |
| Quintile                                                                 | NYSE   | EW      | 0.24<br>[1.87]     | 0.28<br>[2.17]           | 0.26<br>[2.04]          | 0.20<br>[1.58]          |                         |                         |
| Quintile                                                                 | Name   | VW      | 0.44<br>[3.46]     | 0.38<br>[2.99]           | 0.50<br>[4.13]          | 0.49<br>[4.05]          | 0.38<br>[3.39]          | 0.36<br>[3.30]          |
| Quintile                                                                 | Cap    | VW      | 0.27<br>[2.46]     | 0.19<br>[1.81]           | 0.31<br>[3.11]          | 0.31<br>[3.09]          | 0.28<br>[3.06]          | 0.27<br>[2.97]          |
| Decile                                                                   | NYSE   | VW      | 0.47<br>[3.31]     | 0.42<br>[2.92]           | 0.58<br>[4.39]          | 0.60<br>[4.57]          | 0.50<br>[4.00]          | 0.51<br>[4.12]          |



**Table 3:** Conditional sort on size and DTF

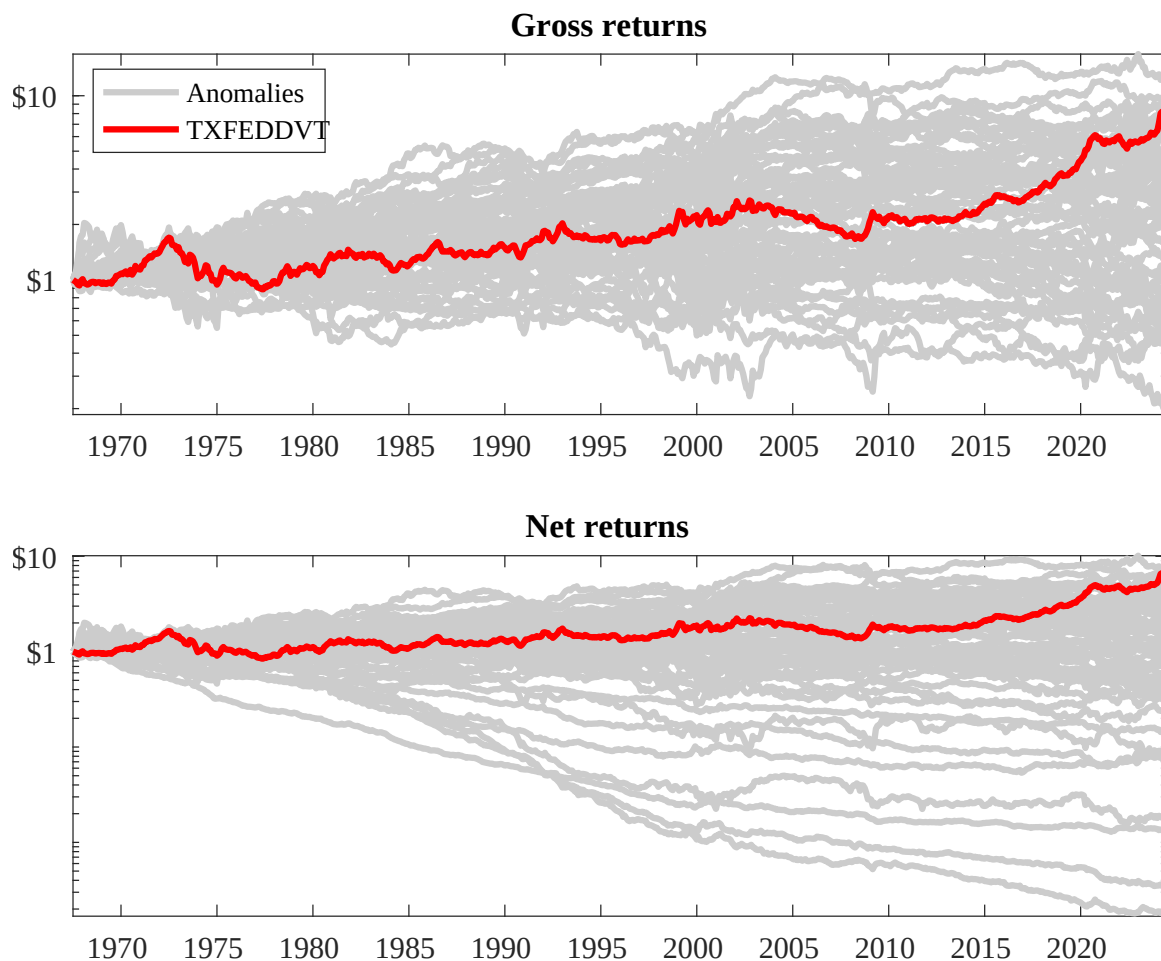
This table presents results for conditional double sorts on size and DTF. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on DTF. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high DTF and short stocks with low DTF. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196706 to 202406.

| Panel A: portfolio average returns and time-series regression results |               |                |                |                |                |                |                                                    |                 |                |                |                  |                  |
|-----------------------------------------------------------------------|---------------|----------------|----------------|----------------|----------------|----------------|----------------------------------------------------|-----------------|----------------|----------------|------------------|------------------|
| Size quintiles                                                        | DTF Quintiles |                |                |                |                | DTF Strategies |                                                    |                 |                |                |                  |                  |
|                                                                       |               | (L)            | (2)            | (3)            | (4)            | (H)            | $r^e$                                              | $\alpha_{CAPM}$ | $\alpha_{FF3}$ | $\alpha_{FF4}$ | $\alpha_{FF5}$   | $\alpha_{FF6}$   |
|                                                                       | (1)           | 0.73<br>[2.51] | 0.35<br>[1.13] | 0.56<br>[2.45] | 1.09<br>[3.74] | 0.89<br>[3.74] | 0.16<br>[0.89]                                     | 0.20<br>[1.12]  | 0.15<br>[0.84] | 0.19<br>[1.05] | -0.13<br>[-0.75] | -0.08<br>[-0.48] |
|                                                                       | (2)           | 0.48<br>[1.94] | 0.73<br>[3.37] | 0.79<br>[3.85] | 0.79<br>[3.73] | 0.78<br>[3.31] | 0.30<br>[2.33]                                     | 0.34<br>[2.54]  | 0.34<br>[2.58] | 0.31<br>[2.31] | 0.10<br>[0.80]   | 0.08<br>[0.67]   |
|                                                                       | (3)           | 0.70<br>[3.07] | 0.61<br>[3.23] | 0.73<br>[3.90] | 0.94<br>[4.66] | 0.81<br>[3.65] | 0.10<br>[0.78]                                     | 0.12<br>[0.93]  | 0.15<br>[1.11] | 0.18<br>[1.29] | -0.15<br>[-1.27] | -0.11<br>[-0.93] |
|                                                                       | (4)           | 0.60<br>[2.90] | 0.61<br>[3.41] | 0.68<br>[3.69] | 0.71<br>[3.72] | 0.79<br>[3.65] | 0.18<br>[1.50]                                     | 0.15<br>[1.19]  | 0.22<br>[1.81] | 0.19<br>[1.56] | 0.02<br>[0.20]   | 0.01<br>[0.07]   |
|                                                                       | (5)           | 0.45<br>[2.61] | 0.49<br>[3.17] | 0.53<br>[3.14] | 0.62<br>[3.45] | 0.69<br>[3.34] | 0.24<br>[1.82]                                     | 0.12<br>[0.98]  | 0.27<br>[2.30] | 0.28<br>[2.34] | 0.30<br>[2.67]   | 0.30<br>[2.60]   |
| Panel B: Portfolio average number of firms and market capitalization  |               |                |                |                |                |                |                                                    |                 |                |                |                  |                  |
| Size quintiles                                                        | DTF Quintiles |                |                |                |                | DTF Quintiles  |                                                    |                 |                |                |                  |                  |
|                                                                       |               | Average $n$    |                |                |                |                | Average market capitalization (\$10 <sup>6</sup> ) |                 |                |                |                  |                  |
|                                                                       |               | (L)            | (2)            | (3)            | (4)            | (H)            | (L)                                                | (2)             | (3)            | (4)            | (H)              |                  |
|                                                                       | (1)           | 116            | 115            | 118            | 119            | 119            | 8                                                  | 7               | 10             | 11             | 11               |                  |
|                                                                       | (2)           | 49             | 49             | 49             | 49             | 49             | 21                                                 | 22              | 22             | 22             | 23               |                  |
|                                                                       | (3)           | 42             | 42             | 42             | 42             | 42             | 47                                                 | 48              | 48             | 48             | 48               |                  |
|                                                                       | (4)           | 41             | 41             | 41             | 41             | 41             | 124                                                | 125             | 127            | 125            | 123              |                  |
| (5)                                                                   | 44            | 44             | 44             | 44             | 44             | 959            | 1399                                               | 1212            | 1392           | 981            |                  |                  |



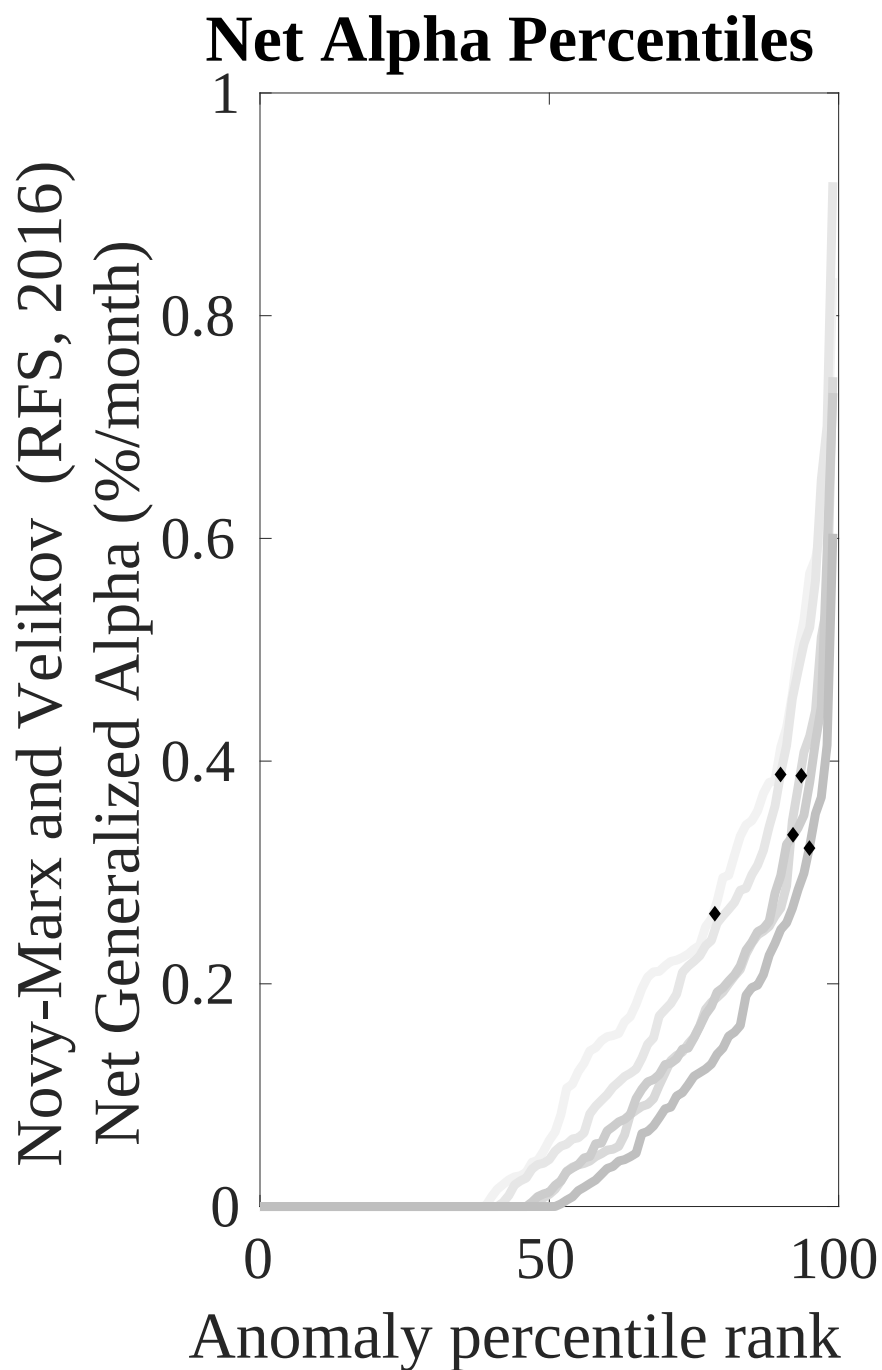
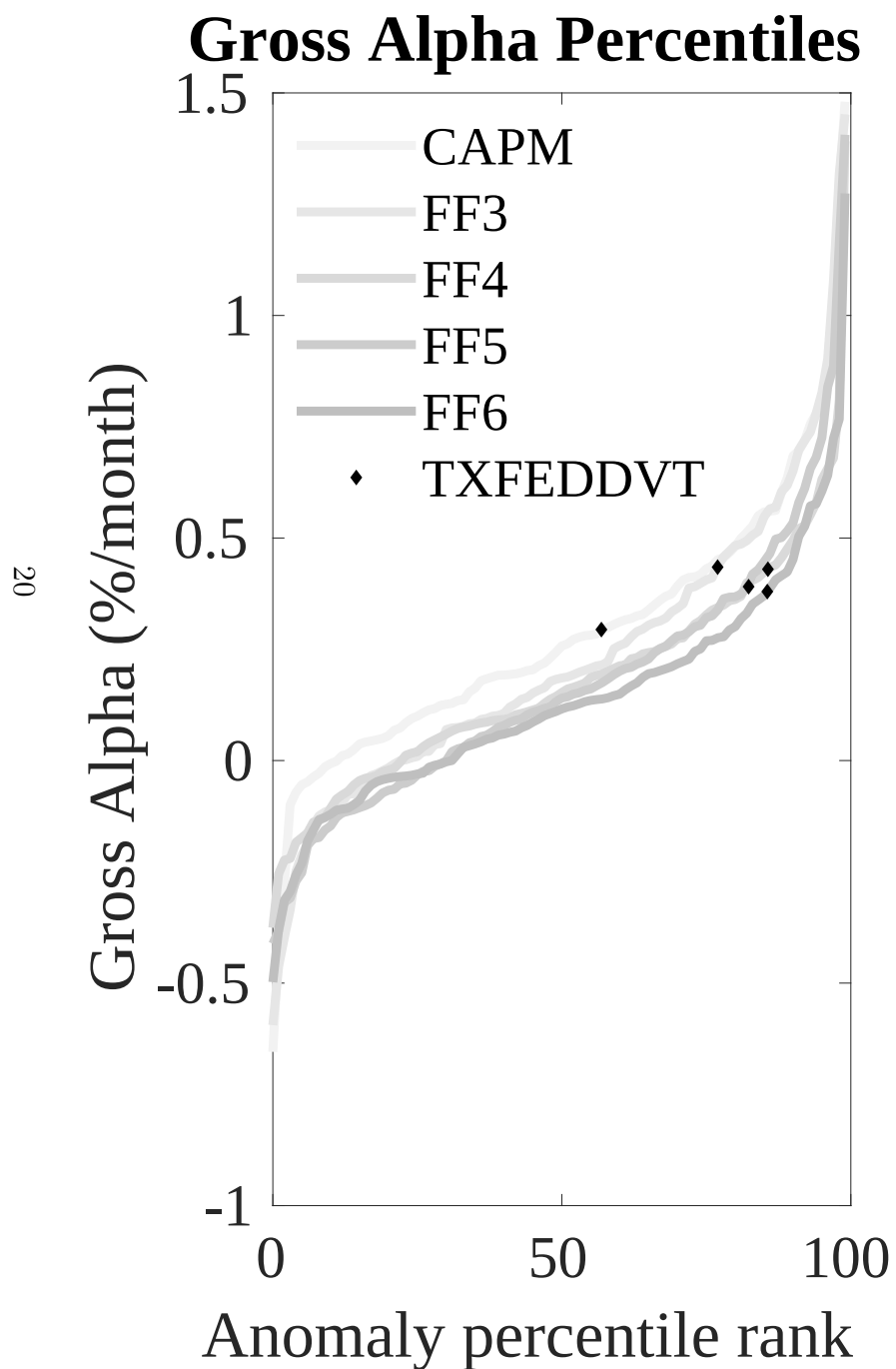
**Figure 2:** Distribution of Sharpe ratios.

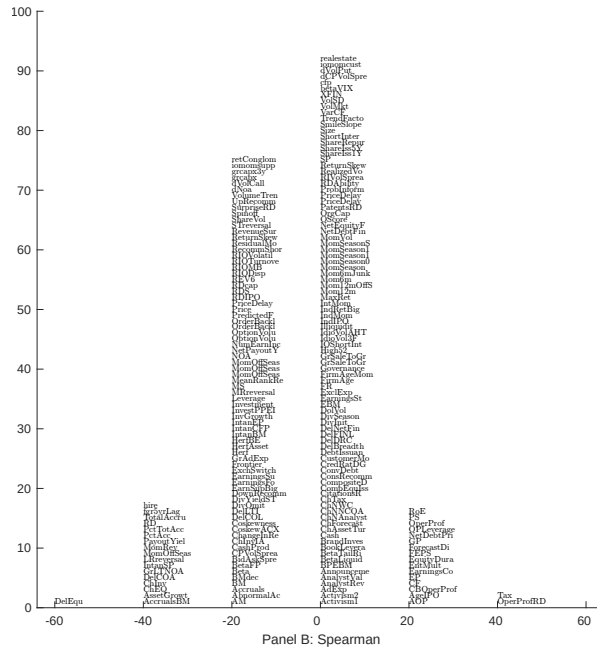
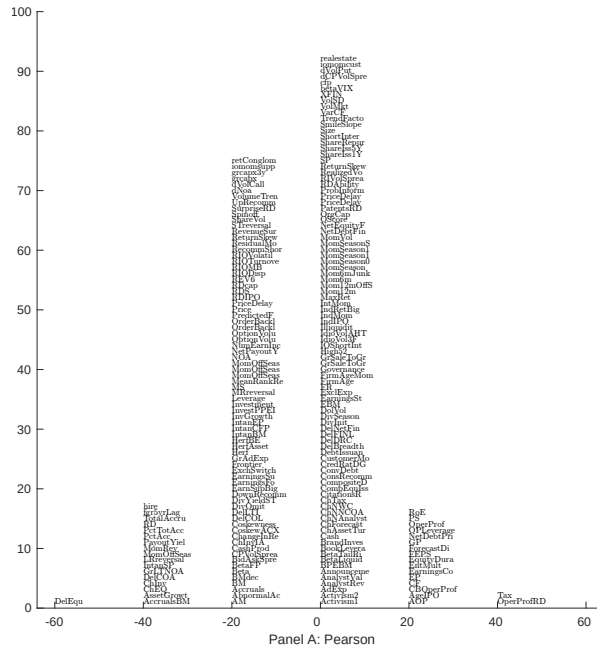
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the DTF with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.



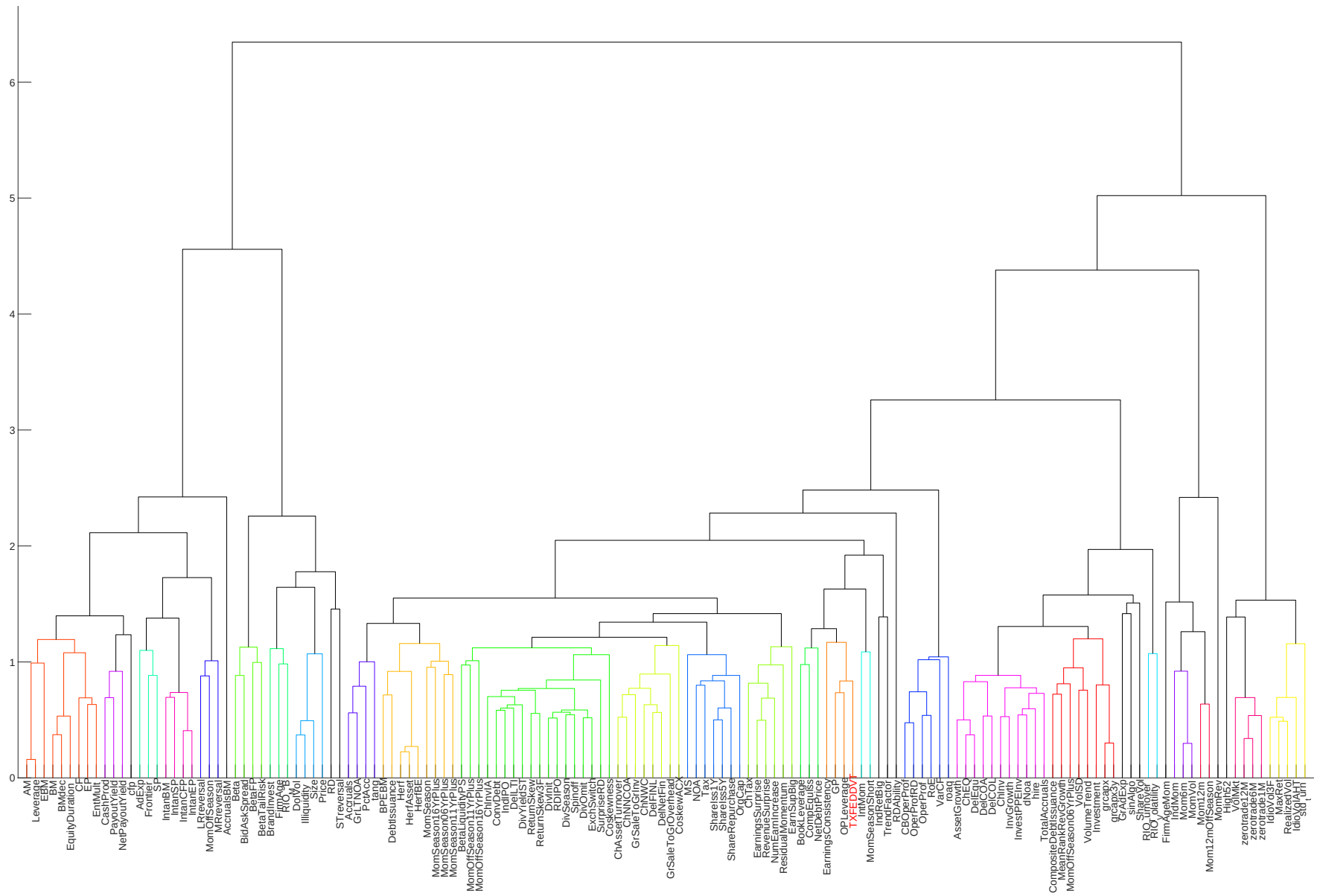
**Figure 3:** Dollar invested.

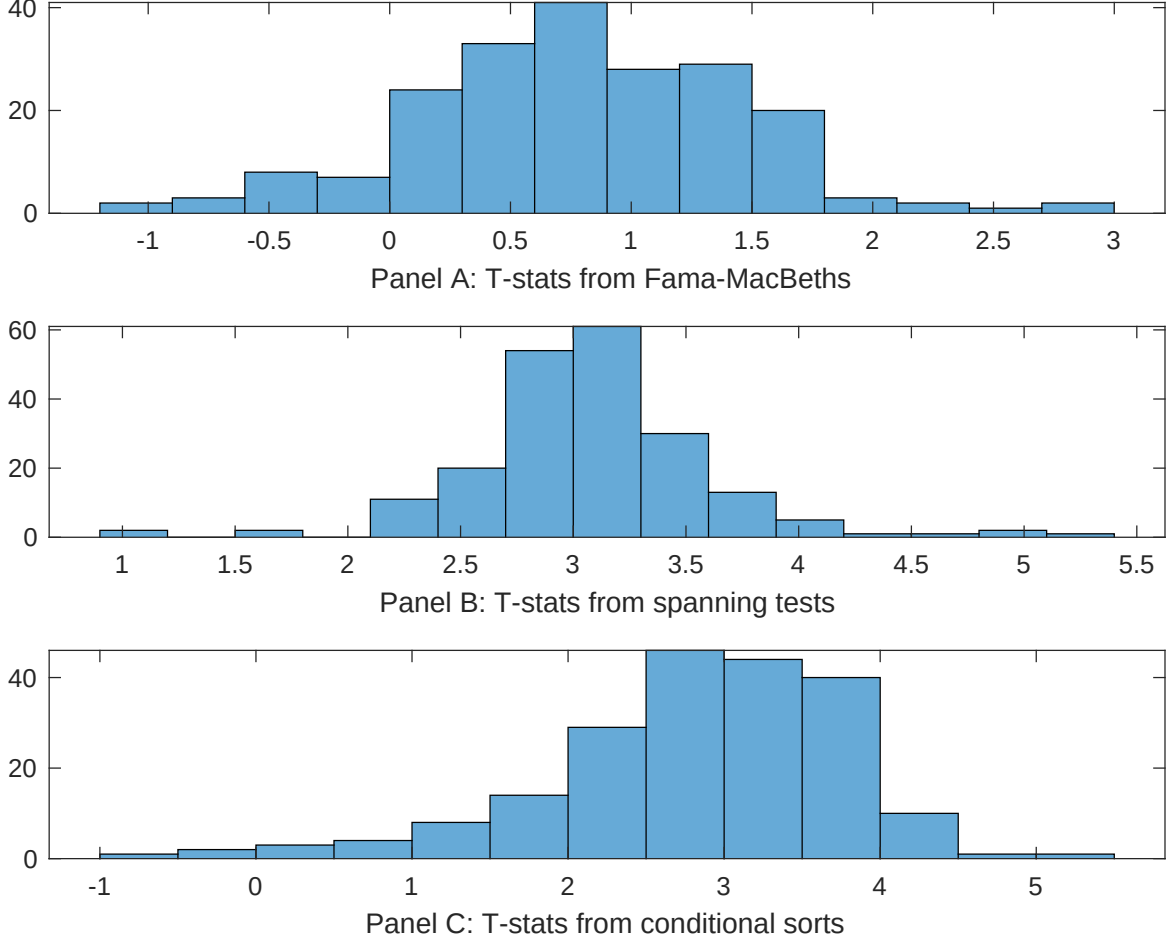
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the DTF trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.





**Figure 5:** Distribution of correlations. This figure plots a name histogram of correlations of 203 filtered anomaly signals with DTF. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.





**Figure 7:** Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of DTF conditioning on each of the 203 filtered anomaly signals one at a time. Panel A reports t-statistics on  $\beta_{DTF}$  from Fama-MacBeth regressions of the form  $r_{i,t} = \alpha + \beta_{DTF}DTF_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$ , where  $X$  stands for one of the 203 filtered anomaly signals at a time. Panel B plots t-statistics on  $\alpha$  from spanning tests of the form:  $r_{DTF,t} = \alpha + \beta r_{X,t} + \epsilon_t$ , where  $r_{X,t}$  stands for the returns to one of the 203 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 203 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on DTF. Stocks are finally grouped into five DTF portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DTF trading strategies conditioned on each of the 203 filtered anomalies.

**Table 4:** Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on DTF, and the six most closely related anomalies. The regressions take the following form:  $r_{i,t} = \alpha + \beta_{DTF}DTF_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$ . The six most closely related anomalies,  $X$ , are gross profits / total assets, Growth in book equity, Change in equity to assets, Payout Yield, Asset growth, Operating leverage. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

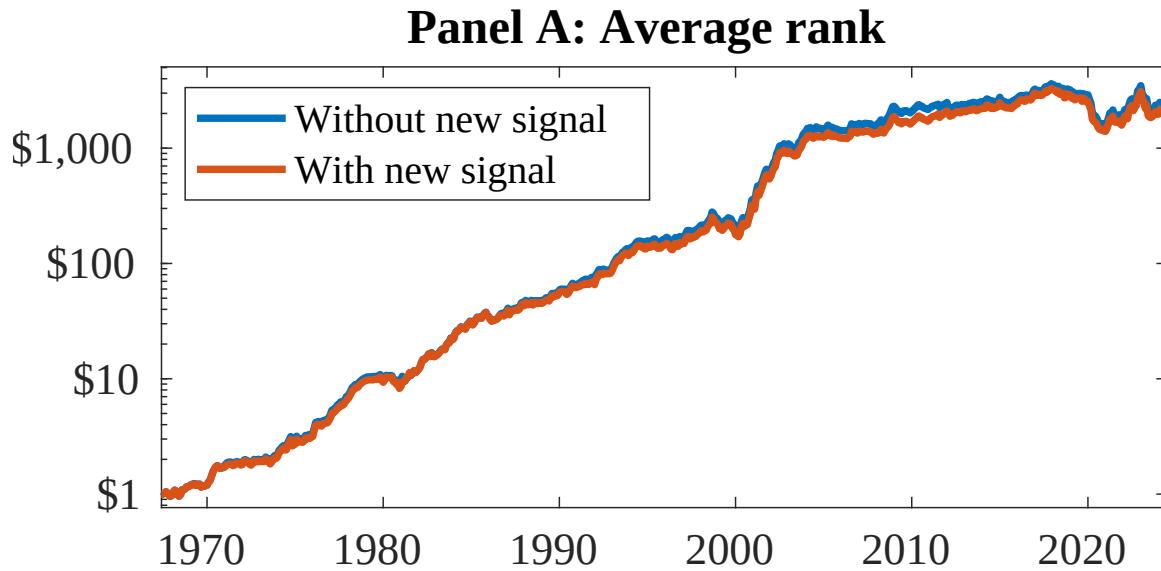
|                 |                |                |                |                  |                |                |                  |
|-----------------|----------------|----------------|----------------|------------------|----------------|----------------|------------------|
| Intercept       | 0.85<br>[3.97] | 0.17<br>[6.34] | 0.11<br>[5.74] | 0.11<br>[5.88]   | 0.12<br>[6.17] | 0.95<br>[4.80] | 0.12<br>[2.68]   |
| DTF             | 0.64<br>[0.32] | 0.25<br>[1.60] | 0.27<br>[1.71] | 0.32<br>[2.30]   | 0.25<br>[1.52] | 0.17<br>[1.01] | 0.35<br>[2.11]   |
| Anomaly 1       | 0.79<br>[2.08] |                |                |                  |                |                | 0.29<br>[0.81]   |
| Anomaly 2       |                | 0.53<br>[2.80] |                |                  |                |                | 0.19<br>[0.44]   |
| Anomaly 3       |                |                | 0.17<br>[2.81] |                  |                |                | -0.21<br>[-0.13] |
| Anomaly 4       |                |                |                | -0.47<br>[-1.25] |                |                | -0.55<br>[-1.46] |
| Anomaly 5       |                |                |                |                  | 0.96<br>[6.76] |                | 0.73<br>[5.47]   |
| Anomaly 6       |                |                |                |                  |                | 0.18<br>[1.96] | 0.49<br>[0.67]   |
| # months        | 684            | 684            | 684            | 668              | 684            | 684            | 668              |
| $\bar{R}^2(\%)$ | 1              | 1              | 1              | 1                | 1              | 1              | 0                |



**Table 5:** Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the DTF trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form:  $r_t^{DTF} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$ , where  $X_k$  indicates each of the six most-closely related anomalies and  $f_j$  indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies,  $X$ , are gross profits / total assets, Growth in book equity, Change in equity to assets, Payout Yield, Asset growth, Operating leverage. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

|                 |                   |                   |                   |                   |                   |                   |                   |
|-----------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| Intercept       | 0.27<br>[2.67]    | 0.39<br>[3.75]    | 0.38<br>[3.62]    | 0.37<br>[3.57]    | 0.38<br>[3.62]    | 0.34<br>[3.35]    | 0.28<br>[2.94]    |
| Anomaly 1       | 34.83<br>[8.54]   |                   |                   |                   |                   |                   | 25.96<br>[5.86]   |
| Anomaly 2       |                   | -22.87<br>[-3.98] |                   |                   |                   |                   | -55.03<br>[-7.22] |
| Anomaly 3       |                   |                   | -2.12<br>[-0.40]  |                   |                   |                   | 40.53<br>[5.44]   |
| Anomaly 4       |                   |                   |                   | -14.64<br>[-3.92] |                   |                   | -4.41<br>[-1.10]  |
| Anomaly 5       |                   |                   |                   |                   | -0.19<br>[-0.03]  |                   | 5.17<br>[0.77]    |
| Anomaly 6       |                   |                   |                   |                   |                   | 31.98<br>[7.87]   | 25.92<br>[6.14]   |
| mkt             | 4.44<br>[1.89]    | 4.29<br>[1.75]    | 5.08<br>[2.05]    | 2.11<br>[0.84]    | 5.04<br>[2.03]    | 5.45<br>[2.30]    | 1.31<br>[0.56]    |
| smb             | 14.83<br>[4.28]   | 16.98<br>[4.71]   | 16.37<br>[4.49]   | 15.69<br>[4.36]   | 16.37<br>[4.45]   | 2.60<br>[0.67]    | 4.77<br>[1.27]    |
| hml             | -2.61<br>[-0.53]  | -16.23<br>[-3.37] | -18.53<br>[-3.80] | -11.53<br>[-2.26] | -18.79<br>[-3.89] | -0.53<br>[-0.10]  | 10.85<br>[2.13]   |
| rmw             | 26.30<br>[5.09]   | 45.28<br>[9.43]   | 46.19<br>[9.48]   | 45.10<br>[9.45]   | 46.39<br>[9.57]   | 28.55<br>[5.53]   | 17.44<br>[3.40]   |
| cma             | -48.45<br>[-7.10] | -29.66<br>[-3.27] | -50.08<br>[-5.58] | -44.56<br>[-6.12] | -52.02<br>[-5.02] | -59.90<br>[-8.64] | -46.07<br>[-4.60] |
| umd             | 1.01<br>[0.43]    | 1.97<br>[0.81]    | 1.48<br>[0.60]    | -0.88<br>[-0.35]  | 1.52<br>[0.61]    | 1.36<br>[0.57]    | 2.30<br>[1.00]    |
| # months        | 684               | 684               | 684               | 680               | 684               | 684               | 680               |
| $\bar{R}^2(\%)$ | 40                | 35                | 34                | 35                | 34                | 39                | 46                |



**Figure 8:** Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 156 anomalies. The red solid lines indicate combination trading strategies that utilize the 156 anomalies as well as DTF. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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